

B. Tech. 1st Year (Common to all Branches of Engineering and Technology)

3700

10/

ME10149: ENGINEERING GRAPHICS

CODE	SUBJECT NAME	Hours per Week			Credits			Maximum Marks				
								Theory		Practical		Total
		L	T	P	T	P	Total	Th.	CW	SW	Pr.	
ME 10149	Engineering Graphics	3	-	4	2	2	4	70	30	40	60	200

PRE-REQUISITES: Nil

Course Objectives

1. To familiarize the student with- drawing instruments, engineering scales and curves.
2. To understand the projections of 1D, 2D and 3D elements using orthographic and isometric views.
3. To familiarize him/ her with the sectioning of solids and development of surfaces.
4. To familiarize him with modern drafting/designing tools and learn to communicate through it.

Course Outcomes: After completion of this course students will be able to

CO 1	Distinguish standard drawing conventions, draw curves and scales using drawing instruments
CO 2	Apply the concept of orthographic projection of lines and planes.
CO 3	To draw the projections of various solids inclined to both planes and its sections and true shape.
CO 4	Differentiate development of surfaces and concept of isometric projection.
CO 5	To draw Orthographic Projections of simple objects and machine parts using drafting tools and software's for creating 2-D and 3-D shapes.

COURSE CONTENTS

THEORY

Unit 1

Basic Concepts, Scales and Curves: Principles of Engineering Graphics and their significance, usage of Drawing instruments, lines, lettering and dimensioning. Scales- Plain, Diagonal Scales, Conic sections including the Rectangular Hyperbola. Cycloidal curves, Involute and Archimedean Spiral.

Unit 2

Projections of Line and Planes: Principles of Orthographic Projections- Conventions, Projection of points and lines inclined to both planes, HT, VT. Projections of planes, inclined Planes -Auxiliary Planes AIP and AVP.

Unit 3

Projection of Solids: Projections of Right regular Solids inclined to both the planes. Auxiliary Views
Sections of Solids: Sections of Geometrical Solids, True shape.

Unit 4

Development of Surfaces: Development of surfaces of Right Regular Solids- Prism, Pyramid, Cylinder and Cone

Isometric Projections: Principles of isometric projection- Isometric Scale. Isometric Views and projections of Lines, Planes and circles: Conversion of Orthographic Views of Simple and compound Solids brackets and simple machine parts into isometric view and projections.

Unit 5

Orthographic Projections: Conversion of isometric Views to Orthographic Views of simple machine parts.

Introduction to Drafting Software: Use and applications of Computer aided drafting (CAD) tools.

ASSESSMENT:

The following methods shall be adopted for the assessment of this course;

1. **Theory Examination** (70 marks) on the basis of end term theory paper examination (from Units 1 to 5)

2. **Class Work** (30 marks) on the basis of regular evaluation of assignments, two mid semester tests and class attendance.
3. **Sessional Work** (40 marks) on the basis of assignments, internal viva and attendance. Student are required to submit at least six imperial drawing sheets/ computer aided drawing print.
4. **Practical examination** (60 marks) on the basis of evaluating practical knowledge, Quiz and viva voce (From Unit 1 to 5).

TEXT BOOKS RECOMMENDED:

1. N.D. Bhatt and V.M. Panchal, Engineering Drawing Plane and Solid Geometry, Charotar Publishing House.

REFERENCE BOOKS:

1. Agrawal Basant and Agrawal C.M. (2018), Engineering Drawing, McGraw Hill Publishing.
2. Shah, MB & Rana BC (2008), Engineering Drawing and Computer Graphics; Pearson Education.
3. Narayana, K.L. & P Kannaiah (2008), Text book on Engineering Drawing, Scitech Publishers
4. (Corresponding set of) CAD Software Theory and use

B. TECH. I YEAR (4YDC) COMMON FOR ALL BRANCHES

MA 10011: MATHEMATICS – I

HOURS PER WEEK			CREDITS			MAXIMUM MARKS				
L	P	Tu	Th	P	Tu	THEORY		PRACTICAL		TOTAL MARKS
						CW	Th.	SW	Pr.	
3	--	1	4	--	--	30	70	--	--	100

PRE –REQUISITES: NIL

COURSE OBJECTIVES

Enable the students to apply the knowledge of Mathematics in various engineering fields by making them

1. To revise the basic concepts of differential calculus and their applications for functions of several variables in engineering problems.
2. To incorporate the knowledge of multiple integral to support the concurrent and subsequent engineering studies.
3. To develop the basic concepts of measure of central tendency, measure of dispersion, correlation and curve fitting.

COURSE OUTCOMES

After completing this course student will be able to

- CO1:** Develop competency in applying the ideas of partial derivatives and use Taylor's and Maclaurin's series for series expansions of function and obtain approximate values.
- CO2:** Apply the knowledge of Lagrange's multipliers for finding the extreme values of functions and understand the concept of curvature and asymptotes.
- CO3:** Evaluate the integral in terms of beta and gamma functions and formulate multiple integrals.
- CO4:** solve application problems of area, length and volume of various types of curve.
- CO5:** Analyze statistical data using measures of central tendency, measure of dispersion and acquire the knowledge of correlation and curve fitting.

COURSE CONTENTS

THEORY

- Unit 1** **Differential Calculus:** Partial derivatives: Definition, Euler's theorem of homogeneous function, Differentiation of implicit function, Total differential coefficients, Jacobians, Expansion of functions by Taylor's and Maclaurin's series of one and two variables .

- Unit 2 Differential Calculus (Continued):** Maxima and Minima of functions of two variables, Lagrange's method of undetermined multipliers and their applications, Curvature in Cartesian and polar coordinates, Asymptotes (cartesian coordinates).
- Unit 3 Integral Calculus:** Beta and Gamma functions, Elementary ideas of multiple integrals, Change of order of integration and change of variables in double integrals using Jacobians.
- Unit 4 Integral Calculus (Continued):** Detailed study of tracing of curves, Area, Length of curve, Volume and Surface of revolution.
- Unit 5 Statistics:** Review of measures of Central Tendencies and measure of Dispersion, Correlation : Karl Pearson and Spearman's rank correlation, Curve fitting by the method of least squares, fitting of straight line, second degree parabolas and more general curves.

ASSESSMENT

1. Internal Assessment for continuous evaluation, mid-term tests, tutorials, class performance, etc. (30%) .
2. End semester Theory Exam (70%) .

TEXT BOOKS RECOMMENDED

1. B. V. Ramana, Higher Engineering Mathematics, Tata McGraw Hill Publishing Company Ltd., New Delhi, 2006.
2. H. K. Das, Advanced Engineering Mathematics, S. Chand and Company Ltd., New-Delhi, 2009.

REFERENCE BOOKS

1. S. S. Sastry, Engineering Mathematics Volume I and Volume II, PHI Learning Private Limited, New Delhi 2008.
2. Tarit Majumdar , Engineering Mathematics Volume – I , New central Book Agency (P) LTD, Kolkata.
3. S Pal and S. C. Bhunia, Engineering Mathematics, Oxford University Press, New Delhi, 2015.
4. Probability and Statistics, Murray R. Spiegel, John J. Schiller, R. Alu Srinivasan, Schaum's Outline Series McGraw Hill, 2009, Third Edition.

PH10017: PHYSICS / भौतिकी

Subject Code / विषय संकेत	Classes/कक्षाएं			Maximum Marks/अधिकतम अंक					Credits / आकलन		
	L	T	P	CW	End	SW	End	Total	T	P	Tot
PH10017	2	1	0	30	70	-	-	100	3	0	3
PH10151	0	0	2	-	-	20	30	50	0	1	1

COURSE OBJECTIVES / अध्ययन के उद्देश्य

- CO# 1 To provide knowledge and understanding of basic, applied, and modern physics. / मूल, व्यावहारिक और आधुनिक भौतिकी का ज्ञान और समझ क्षमता प्रदान करना।
- CO# 2 To generate attitude and interest to solve problems at macro, and micro level systems. / स्थूल, सूक्ष्म व नैनो स्तर की प्रणालियों में प्रश्नों को हल करने के लिए दृष्टिकोण और रुचि उत्पन्न करना।
- CO# 3 To understand the classification and properties of both basic and applied physics at nanoscale systems / मूल और अनुप्रयुक्त भौतिकी दोनों के वर्गीकरण और गुणों नैनो स्तर पर समझना।
- CO# 4 To update the knowledge of physics tools, instruments, and techniques incorporating human values and safety measures. / मानव मूल्यों और सुरक्षा उपायों को शामिल करते हुए भौतिकी के उपकरणों और तकनीकों के ज्ञान को सामयिक बनाना।
- CO# 5 To identify, conduct, formulate and solve engineering problems with the applied knowledge of Physics. / भौतिकी के मूल और व्यावहारिक ज्ञान के साथ अभियंत्रिकी प्रश्नों की पहचान, संचालन, सूत्रीकरण और समाधान करना।

PH10017: PHYSICS/ भौतिकी COURSE CONTENTS

/ अध्ययन की विषयवस्तु

- Unit-1 Optics:** Theory of superposition, sustained interference, division of wavefront and amplitude, Newton's rings, Classes of diffraction, diffraction at N slits, Rayleigh's criteria and resolving power.।

प्रकाशिकी: अध्यरोपण का सिद्धांत, स्थिर व्यतिकरण के लिए शर्तें, तरंगग्र और आयाम का विभाजन, न्यूटन वलय, विवर्तन के प्रकार, N (ग्रेटिंग) रेखाछिद्रों द्वारा विवर्तन, रैले के मानदंड और विभेदन क्षमता।

- Unit-2 Laser & Fiber Optics:** Photon absorption & emission, components of lasers, optical resonator, Einstein's A & B coefficients, Population inversion, Ruby and He-Ne lasers, applications. Classification of fibers, acceptance angle, numerical aperture, V-number, losses, FO sensors, FO communication Systems.

लेजर एवं तंतु (फाइबर) प्रकाशिकी: तात्क्षणिक और उद्दीप्त उत्सर्जन, लेजर के घटक, प्रकाशीय अनुनादक, आइंस्टीन के A और B गुणांक, समष्टि प्रतिरोमन, रूबी और He-Ne लेजर, लेजर के अनुप्रयोग। तंतु (फाइबर) के वर्गीकरण, स्वीकृति कोण, संख्यात्मक छिद्र, V-नंबर, क्षोणन, प्रकाशीय तंतु सुग्राही, प्रकाशीय तंतु संचार प्रणाली।

- Unit-3 Special Theory of relativity (STR):** Postulate of STR, time dilation, length contraction, twin paradox, Relativistic

velocity of particles, Relativistic variation of mass, mass-energy relation.

विशिश्ट आपेक्षिकता सिद्धांत: आपेक्षिकता सिद्धांत, समय फैलाव, लंबाई संकुचन, बुद्धवा विरोधाभास, कणों का आपेक्षिक वेग, द्रव्यमान का आपेक्षिक परिवर्तन, द्रव्यमान-ऊर्जा संबंध।

- Unit-4 Quantum Theory:** Planck's radiation formula, Ultraviolet catastrophe, Compton's effect, de Broglie's and Heisenberg's hypotheses, Schrodinger's wave equation, interpretation of wave function, particle in a 1-D potential well.

क्वांटम सिद्धांत: प्लैंक का विकिरण सूत्र, पराबैंगनी विपात, कॉम्पटन का प्रभाव, डी ब्रोग्ली की पदार्थ तरंगों की अवधारणा, हाइजेनबर्ग का अनिश्चितता सिद्धांत, श्रोडिंगर का तरंग समीकरण, तरंग फलन की भौतिक व्याख्या, एक-आयामी विभव कूप में कण की परिकल्पना।

- Unit-5 Thermodynamics:** Thermodynamic properties and processes, heat and work. Zeroth law and First law of thermodynamics. Steady flow energy equation, basic concepts of Second law of thermodynamics.

थर्मोडायनामिक्स: ऊष्मागतिकी गुण और प्रक्रियाएँ, ऊष्मा और कार्य, ऊष्मागतिकी का त्रिरोध और पहला नियम, स्थिर प्रवाह ऊर्जा समीकरण, ऊष्मागतिकी के दूसरे नियम की बुनियादी अवधारणाएँ।

PH10151: PHYSICS/ भौतिकी - LIST OF

EXPERIMENTS / प्रयोगों की सूची

1. Error Analysis in Physics Laboratory. / भौतिकी प्रयोगशाला में त्रुटि विश्लेषण।
2. To determine the wavelength of sodium light by Newton's rings method. / न्यूटन वलय विधि द्वारा सोडियम प्रकाश की तरंगदैर्घ्य ज्ञात करना।
3. To study the variation of magnetic field along the axis of a circular coil carrying current and to calculate the radius of the coil. / एक वृत्ताकार कुंडली के अक्ष के अनुदिश चुंबकीय क्षेत्र की भिन्नता का अध्ययन करना तथा कुंडली की विज्या की गणना करना।
4. To measure the numerical aperture of given optical fiber. / दिए गए ऑप्टिकल फाइबर के संख्यात्मक छिद्र की गणना करना।
5. To measure Planck's constant using light emitting diodes (LED) of various colors and to understand work function. / विभिन्न रंगों के प्रकाश उत्सर्जक डायोड (LED) का उपयोग करके प्लैंक के नियतांक को मापना और कार्यफलन को समझना।
5. To study working of laser using PhET Module./PhET मॉड्यूल का उपयोग करके लेजर की कार्यप्रणाली का अध्ययन करना।
7. To study the law of dispersion by a triangular prism using raytrace. / त्रैभुजाकार प्रिज्म द्वारा प्रकीर्णन का अध्ययन करना।

8. To understand and confirm Heisenberg's uncertainty principle using single slit diffraction. / एकल रेखाछिद्र विवर्तन का उपयोग करते हुए हाइजेनबर्ग के अनिश्चितता सिद्धांत को समझना और पुष्टि करना।
9. To determine the wavelength of prominent spectral lines of mercury light by a plane transmission grating using normal incidence. / अभिलंब आपतन का उपयोग करते हुए एक समतल पारगमन ग्रेटिंग द्वारा प्रकाश की प्रमुख वर्णक्रमीय रेखाओं की तरंग दैर्घ्य निर्धारित करना।
10. To measure the charge to mass ratio of an electron using Thomson method and to find the polarity of the charge of electron. / थॉमसन विधि का उपयोग करके एक इलेक्ट्रॉन के आवेश और द्रव्यमान के अनुपात को मापना तथा इलेक्ट्रॉन के आवेश की ध्रुवता ज्ञात करना।
11. To verify Kirchhoff's law and Inverse-square law in thermodynamics using Leslie's cube. / लेस्ली के क्यूब का उपयोग करके, ऊष्मागतिकी के किर्चॉफ के नियम और व्युत्क्रम-वर्ग नियम को सत्यापित करना।

COURSE OUTCOME / अध्ययन के निष्कर्ष

By the study of one-semester course, the students will be armed with the / एक अर्ध वार्षिक सत्र के पाठ्यक्रम के अध्ययन से, छात्र सक्षम होंगे

- CO# 1 knowledge of multiphysics to understand and solve basic engineering problems. / बहु-भौतिकी का ज्ञान तथा मूल अभियांत्रिकी प्रश्नों को समझने और हल करने में
- CO# 2 logical attitude towards the solution of engineering problems with modern physics implementation. / आधुनिक भौतिकी कार्यान्वयन के साथ अभियांत्रिकी प्रश्नों के समाधान के लिए तार्किक दृष्टिकोण में
- CO# 3 ability to use modern techniques and tools including software involving advanced physics to engineering subjects / अभियांत्रिकी विषयों के लिए उन्नत भौतिकी से जुड़े सॉफ्टवेयर सहित आधुनिक तकनीकों और उपकरणों का उपयोग करने की क्षमता में।
- CO# 4 comprehensive applications of engineering physics of optics, laser & fiber optics theory of relativity, quantum mechanics, thermodynamics. / प्रकाशिकी, लेजर और फाइबर ऑप्टिक्स, सापेक्षता के सिद्धांत, न्वांटम यांत्रिकी, ऊष्मागतिकी के इंजीनियरिंग भौतिकी के व्यापक अनुप्रयोग।

- CO# 5** creative engineering solutions for incorporating human values and measures with the applied knowledge. / अनुप्रयुक्त भौतिकी ज्ञान के साथ मानवीय मूल्यों और सुरक्षा उपायों को शामिल करते हुए समाज के लिए रचनात्मक अभियांत्रिकी समाधान करने में।

TEXT BOOKS / पाठ्य पुस्तक

1. N. Subramanyam and B. Lal: A Text book of Optics, (S. Chand, New Delhi) 2010.
2. A. Beiser, S. Mahajan and S. R. Choudhary: Concepts of Modern Physics, 6th Ed., (Tata McGraw Hill, New Delhi) 2012.
3. A. Ghatak: Optics, 4th Edition, (Tata McGraw-Hill, New Delhi) 2009.

REFERENCE BOOKS / संदर्भ पुस्तक

1. H. K. Malik and A. K. Singh: Engineering Physics (Tata McGraw Hill New Delhi) 2010.
2. R. P. Feynman, R.B. Leighton and M. Sands: Feynman Lectures on Physics Vol. 1-3, (Addison-Wesley, Delhi) 1995.
3. W. H. Hayt: Engineering Electromagnetic, 5th Ed. (Tata-McGraw Hill, New Delhi) 1995.
4. M. N. O. Sadiku: Elements of Electromagnetic, 3rd Ed. (Oxford Press, New Delhi) 2000.

REFERENCE MANUAL / संदर्भ तालिका

Download Lab Manual from the website (<https://sgsits.ac.in>) of Shri G. S. Institute of Technology and Science, Indore. Consult the references given in each chapter of the manual for more information. Download Tutorial Sheets from <http://dx.doi.org/10.13140/RG.2.2.28342.63044> / श्री जी. एस. प्रौद्योगिकी एवं विज्ञान संस्थान, इंदौर की वेबसाइट (<https://sgsits.ac.in>) से प्रयोगशाला पुस्तिका डाउनलोड करें। अधिक जानकारी के लिए पुस्तिका के प्रत्येक अध्याय में दिए गए संदर्भों को देखें। <http://dx.doi.org/10.13140/RG.2.2.28342.63044> से अनुशिक्षण पत्रक डाउनलोड करें।

ELECTRICAL ENGINEERING DEPARTMENT
B. TECH. FIRST YEAR (4 YDC)
EE10015: FUNDAMENTALS OF ELECTRICAL ENGINEERING (Modified)

HOURS PER WEEK			CREDITS			MAXIMUM MARKS				
T	P	TU	T	P	TU	THEORY		PRACTICAL		TOTAL MARKS
						C W	END SEM	S W	END SEM	
2	-	1	2	-	1	30	70	-	-	100

PRI- REQUISITE: Basic knowledge of science and mathematics at XII level.

COURSE OBJECTIVES:

1. To acquaint students with the basic concepts, elements and properties of electrical circuits and networks.
2. To introduce the concept of phasor and power factor improvement in sinusoidal steady-state analysis.
3. To impart the knowledge of a three-phase system- its generation and analysis.
4. To explain basics of magnetic circuit and its practical aspects applied in transformer. To teach students the basic principles of operation, construction, and application of rotating electrical machines.

COURSE OUTCOMES:

After completing this course, the student will be able to:

EE10015(T)1: Formulate various circuit describing mathematical models.

EE10015(T)2: Utilize ac excited steady state circuit analysis concepts in real life problems such as need of power factor correction, power factor improvement methods and home energy distribution system.

EE10015(T)3: Analyse three phase power generation system and perform basic analysis of three phase star/delta circuits.

EE10015(T)4: Carry out practical magnetic circuit calculations and performance analysis of single phase transformer.

EE10015(T)5: Do performance analysis of various type of motors and develop ability to identify suitable motor for particular application.

COURSE CONTENTS:

THEORY:

UNIT: 1

Voltage and current sources & transformation, resistors, effect of temperature on resistance, capacitors and inductors, their v-i relationships, Kirchhoff's laws, Mesh and Nodal analysis, Star-Delta transformations, Superposition theorem, Thevenin's and Norton's theorem, Maximum power transfer theorem.

UNIT: 2

Circuit analysis in sinusoidal steady state, various signals & waveform analysis, Concept of phasor, impedance, admittance, phasor diagrams, Complex, real and reactive power in ac circuits, Power factor and power factor corrections, capacitor bank, electricity distribution system in houses.

UNIT: 3

Generation of three phase voltages, star and delta connection, Line and Phase quantities, Three phase power, Three phase Power measurement techniques, diesel generator set overview.

UNIT: 4

Magnetic circuits, definitions, B-H curves, Magnetic circuit calculations, laws of electro-magnetic induction, Hysteresis and eddy current losses, single phase transformer construction & principle, EMF equation, equivalent circuit, regulation and efficiency.

UNIT: 5

Basic construction of three phase induction motor, production of rotating magnetic field, principle of working of three phase induction motor, losses, name plate rating, efficiency, type of motor used and rating for domestic and industrial applications.

ASSESSMENT:

- A. Continuous evaluation through two mid-term test with a weightage of 30% of the total marks. It includes class attendance as well as assignments on the course topics.
- B. The end-term theory examination weightage is 70%.

TEXT BOOKS RECOMMENDED:

1. Vincent Del Toro, "Electrical Engineering Fundamentals", Prentice-Hall of India.
2. John Hiley, Keith Brown & Ian McKenzie Smith, (HUGES), "Electrical and Electronic Technology", Pearson Education Ltd.
3. Charles K. Alexander, & Matthew N. O. Sadiku, "Fundamentals of Electric Circuits", McGraw Hill Higher Education.

REFERENCES BOOKS:

1. Fitzgerald Higginbotham & Grabel, "Basic Electrical Engineering", Tata McGraw Hill Publication.
2. Joseph Edminister, Mahmood Nahvi "Electric Circuits", Schaum's outlines series.
3. S. K. Bhattacharya, "Basic Electrical and Electronics Engineering", Pearson India Education Services Pvt. Ltd.
4. I. J. Nagarath, "Basic Electrical Engineering", Tata McGraw Hill Publication.

Proposed Modified Syllabus

w.e.f Sept 2023 onwards

CIVIL ENGINEERING AND APPLIED MECHANICS DEPARTMENT

B. Tech. I YEAR (4YDC)

CE: 10013:

FUNDAMENTALS OF CIVIL ENGINEERING AND APPLIED MECHANICS

CREDITS:

CREDITS:											TOTAL MARKS
PERIOD PER WEEK			CREDITS			MAXIMUM MARKS					
T	P	Tu	T	P	Tu	THEORY		PRACTICAL			
						CW	END SEM	SW	END SEM	100	
2+1	00	01	3	00	00	30	70	00	00		

PRE-REQUISITE: XII standard (HSSC)

COURSE OBJECTIVES:

1. To study various force systems, equilibrium of force system, frictional forces, concept of trusses
2. To learn the concepts of centroids, centers of gravity and moments of inertia of simple and complex geometrical shape. Transmission of power through belt & ropes.
3. To learn the behavior of beams, frames and trusses under different loading conditions and to determine the shear force and bending moment diagram for beams subjected to simple point loads.
4. To study methods and equipments for chaining, linear & angular measurements.
5. To understand the fundamental concepts of leveling.

COURSE OUTCOMES:

The students shall be able to

1. Identify and analyze system of forces, calculate support reactions for beam subjected to various types of loading and determine forces in members of trusses.
2. Determine the Center of Gravity and moment of Inertia of a given Plane Lamina & Solid Bodies.
3. Determine and analyze the Shear force, Bending moment of beams and analyze the trusses and problems related to frictions
4. Understand the working principles of basic equipments used in surveying and measuring distance, perpendicular and oblique offsets from the given reference line.
5. Apply the knowledge of surveying and leveling for computation of area and volume in different operations in civil engineering projects.

COURSE CONTENTS:

THEORY

PART - A: (60% Weightage)
APPLIED MECHANICS

UNIT-1

Forces and Equilibrium: Graphical and Analytical Treatment of Concurrent and non-concurrent Co-planer forces, free body Diagram, Force Diagram and Bow's notations, Application of Equilibrium Concepts: Analysis of plane Trusses: Method of joints, Method of Sections. Frictional force in equilibrium problems.

UNIT-2

Centre of Gravity and Moment of Inertia: Centroid and Centre of Gravity, Moment of Inertia of Area and Mass, Radius of Gyration, Introduction to Product of Inertia and Principle Axes. Power Transmission and its applications.

UNIT-3

Support Reaction and analysis of Beams: Types of Beams: Simply Supported Beam, Overhanging

Beam, and Cantilever Beam. Types of Supports of a Beam or Frame: Roller, Hinged and Fixed Supports. Load on the Beam and Frame: Different Types of Loading. Support Reaction of a Beam or Frame. Introductions to shear force and Bending Moment for Cantilever & simply supported beam with concentrated, distributed load and Couple.

PART - B: (40% Weightage)

SURVEYING

UNIT-4

Surveying & Positioning: Introduction to surveying Instruments –Levels, Theodolites, Plane tables and related devices. Linear & Angular Measurements. Electronic surveying instruments etc. Measurement of distances – conventional and EDM methods, Measurement of directions by different methods.

UNIT-5

Leveling, Mapping & Sensing: Types of Levels Measurement of elevations by different methods. Reciprocal leveling, Mapping details and contouring, Profile Cross sectioning and measurement of areas, volumes, application of measurements in quantity computations, Survey stations, Introduction of remote sensing and its applications.

DEMONSTRATIONS (Surveying Equipments):

OBJECTIVES:

1. To acquire practical knowledge on handling basic survey equipment's.
2. To possess knowledge about chain, compass surveying and open traverse survey.
3. To have the ability to understand leveling and area of irregular figures.

OUTCOMES:-

After demonstrating different instruments and experiments, students shall be able to

1. Understand the working principles of chaining instruments and measuring distance, perpendicular and oblique offsets from the given reference line.
2. Understand the principle of prismatic compass, open traverse survey.
3. Compute the RL of different points on the ground by spot leveling and also compute the area and volume of irregular figure.

List of Demonstration/Experiments:

1. To demonstrate the color/sign conventions and various instruments used in chaining.
2. To demonstrate how to determine the horizontal distance between two points by doing ranging and chaining.
3. To demonstrate measure perpendicular and oblique offsets from a given reference line.
4. To demonstrate prismatic compass and perform open traverse survey.
5. To demonstrate traversing of any regular figure with the help of compass and eliminate local attraction.
6. To demonstrate how to determine RL of different points on the ground by fly differential leveling and compute RL's by height of instrument (HI) method and rise and fall method.
7. To demonstrate how to determine planimeter constants and area of an irregular figure with the help of a planimeter.

ASSESSMENT:

Continuous:

1. Class work will be based on following
 - Submission of assignments based on syllabus of Engineering Mechanics and Surveying.
 - Two midterm tests in a semester and a makeup test if required, Evaluation of Practical calculations & drawing sheets, internal submission and Viva Voice examination by internal examiner during the semester end.
2. Semester-end: Theory examination of 3 Hours duration

RECOMMENDED TEXT BOOKS:

1. Rajput .R.K., *Engineering Mechanics*, Dhanpat Rai and Sons New Delhi.
2. Prasad I. B., *Applied Mechanics*, Khanna Publication New Delhi
3. R. S. Khurmi, *Engineering Mechanics*, , S.Chand Publishing
4. D.S. Bedi, *Engineering Mechanics*, , Khanna Book Publishing Co. (P) Ltd., Delhi
5. R.K. Bansal, *A Textbook of Engineering Mechanics*, , Laxmi Publications
6. Rammamurtham S., *Applied Mechanics*, Dhanpat Rai and Sons New Delhi
7. Punmia, B.C., *Surveying*, Laxmi Publications, New Delhi.
8. Duggal S.K., *Surveying Vol. 1*, Tata McGraw- Hill Education New Delhi

REFERENCE BOOKS:

1. Kumar KL, *Engineering Mechanics*, Tata McGraw- Hill Education New Delhi
2. Ferdinand.P. Beer. E, Russell, Jr Johnston., David Mazurek, Philip J Cornwell, "Vector Mechanics for Engineers: Statics and Dynamics". McGraw – Hill Education New Delhi
3. Timoshenko, and Young D.H., "Engineering Mechanics", Tata Mc-Graw Hill Education New Delhi.
4. Chanchandramouli P.N., *Engineering Mechanics*, PHI Learning Private Limited New Delhi
5. Palanichamy *Basic Civil Engineering*, , McGraw Hill
6. Sathesh Gopi *Basic Civil Engineering*, Pearson Publishers